SDG BLOCKCHAIN ACCELERATOR

Mentor Feedback Report: ClimaFi (Vaka / ClimaFI)

Program: EMURGO × UNDP Blockchain Accelerator (Cohort 2)
Project: Results-Based Payments for Meteorological Data (RBP-MetData)
Geographic Focus: Tanzania (Pilot), Global South (Scale)
Mentorship Focus: Technical Architecture, Regulatory Translation, PoC Design & Verification Logic
Total Mentoring Sessions: 4
Mentor(s): Paul Krugel / Teo Petrovic

1. Initial Positioning and Framing of the Problem

ClimaFi entered the accelerator with a very clearly articulated problem , and more importantly, one that is structurally understood within the UNDP and SOFF ecosystem.

The issue is not a lack of infrastructure. In fact, funding already exists to deploy meteorological stations. The real gap sits in what the team described as the “missing middle”: the inability to continuously verify that these systems are functioning and producing usable data over time.

This framing is important, because it shifts the solution away from:

- hardware deployment
- or generic data platforms

toward something more specific:

> a **verification and accountability layer for results-based climate finance**

From the beginning, the project had a strong conceptual alignment with:

- SOFF’s compliance phase
- WMO standards (GBON/WIS2)
- and UNDP’s push toward results-based financing

The mentoring process therefore did not question the relevance of the solution, but focused on:

> how to translate this institutional logic into a **working system that can be tested without regulatory friction**.

2. A Shift in Understanding: From Payments → Verification

One of the most important evolutions during mentoring was a shift in how the system is positioned.

Initially, the project leaned heavily toward:

- automated payments
- tokenized incentives
- blockchain-based settlement

However, through discussions with stakeholders and technical reviews, it became clear that:

> the immediate problem is not moving money , it is **proving that data is trustworthy**.

This led to a key pivot:

- from a **“payment infrastructure” mindset**
- to a **“digital audit / verification system” mindset**

This shift is already reflected in the design of the **“Data Receipt Tokens”**, which act less like financial instruments and more like:

- cryptographic proofs of performance
- machine-verifiable audit artifacts

This was a critical decision, because it aligns much better with:

- institutional readiness
- regulatory constraints
- and the actual bottleneck in climate finance systems

3. Technical Architecture: Where It’s Strong and Where It’s Sensitive

From a technical standpoint, the system is built around three main layers:

1. Data ingestion and verification (off-chain)
2. Tokenization and audit anchoring (on-chain)
3. Payment logic (shadow / future settlement layer)

This separation is conceptually sound, and in many ways necessary.

The **verification engine** is clearly the core of the system. It is where:

- GBON/WIS2 compliance is enforced
- anomalies are detected
- and data is filtered before it becomes economically meaningful

What became clear during mentoring is that:

> this layer is not just a component , it is the product.

If the verification logic is weak, everything else collapses.
If it is strong, the rest of the system becomes almost optional.

On-chain Layer: Useful but Not the Bottleneck

The blockchain layer (Cardano) is used primarily for:

- token minting (receipt tokens)
- data anchoring (hashes)
- auditability

This is appropriate, but it also became clear that:

> the blockchain is not where most of the technical risk lies.

The system could technically function (for the pilot) with:

- minimal on-chain interaction
- or even delayed anchoring

The real challenge is ensuring:

- deterministic verification
- reproducibility of results
- and consistency between off-chain data and on-chain records

The Critical Dependency: Data Quality and Structure

The hardest part of this system is not writing smart contracts.

It is dealing with:

- inconsistent station data
- latency issues
- missing data points
- connectivity gaps

The introduction of **asynchronous validation** (accepting delayed data) is a strong and necessary adaptation. It reflects a realistic understanding of the environment rather than forcing ideal conditions.

However, this also introduces complexity:

- how far delays are acceptable
- how disputes are handled

- how “quality” is defined vs “availability”

These are not just technical questions , they are:

> **economic and governance questions embedded in code.**

4. PoC Design: One of the Strongest Aspects of the Project

Compared to many teams, ClimaFi stands out in how clearly the PoC is structured.

The phased approach:

- ingestion → verification → tokenization → simulation → evaluation

is not only logical, but also aligned with real-world constraints.

What is particularly well designed is the **“Shadow Payment” model**.

Instead of forcing:

- regulatory approval
- financial flows
- or stablecoin usage

the system creates a **digital twin of the payment process**:

- tokens are minted
- value is calculated
- but no actual funds move

This is a very strong design choice because it:

- de-risks the pilot
- allows institutional stakeholders to engage safely
- and produces measurable outputs (shadow earnings, reports)

It essentially allows the system to answer:

> “Would this work if real money were attached?”
> without actually needing to move money.

5. Where the Real Complexity Lies

A recurring theme across sessions was that the project sits at the intersection of:

- technical systems
- institutional workflows
- and financial mechanisms

And most of the complexity is not purely technical.

For example:

- Who defines what “valid data” is? (WMO vs local reality)
- Who resolves disputes if data is rejected?
- How are ministries incentivized to trust the system?

These questions influence:

- how the verification engine is designed
- how thresholds are set
- how tokens are interpreted

In other words:

> The system is encoding governance into infrastructure.

This is powerful, but also risky if not carefully aligned with stakeholders.

6. What the Team Is Doing Well

The team demonstrates a strong understanding of:

- institutional context (SOFF, UNDP, WMO)
- technical constraints (data quality, latency)
- and adoption risks (regulation, public finance systems)

The decision to:

- start with verification
- introduce shadow payments
- and delay real financial flows

shows a level of maturity that is often missing in similar projects.

There is also a clear effort to make the system:

- interpretable (dashboards, reports)
- not just technically correct

which is critical for stakeholder adoption.

7. Challenges and Risks Going Forward

The main risks are not about feasibility, but about alignment and execution.

One challenge is ensuring that the verification logic:

- remains transparent
- is accepted by all stakeholders
- and does not become a black box

Another is managing expectations.
There is a natural tendency for stakeholders to focus on:

- payments
- funding flows
- and financial outcomes

while the system is still in a validation phase.

There is also a long-term question around:

- how this integrates into government financial systems
- and whether blockchain is visible or abstracted away

8. Final Mentor Perspective

ClimaFi is one of the few projects in the cohort that directly engages with:

- real funding mechanisms
- real institutions
- and real global standards

It is not just a technical solution , it is attempting to redesign how climate finance accountability works.

The strongest part of the project is the recognition that:

trust must be built through **verification first**, not automation.

The PoC is well designed to support that.

If the team continues to:

- prioritize clarity over complexity
- keep the verification layer transparent and defensible
- and maintain close alignment with institutional stakeholders

then the project has a strong chance of moving beyond a pilot and becoming:

a foundational layer for results-based climate finance systems.